Sophia (Zahra) Chehreghan

Urban and spatial planner | Portfolio: https://sophia-chehre.github.io/portfolio/

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Name in official documents: Zahra Chehreghan. I go by Sophia.

Urban planner with a multidisciplinary profile across planning, research, spatial data and programming, and entrepreneurship. Carries projects from defining the problem through data analysis and solutions to interactive results. Current research experience in Germany. Completed MSc in Urban Planning; MSc Spatial Planning in progress.

Research and applied experience

06/2026-present

Student Research Assistant, Geoinformation and Monitoring group | ILS – Research Institute for Regional and Urban Development, Dortmund

Contribute to two urban and regional research projects with teams in Germany.

CATCH4D Upscaling — Student Research Assistant (with Dr. Shaojuan Xu): prepare aerial imagery and LiDAR-derived inputs, run and evaluate deep-learning experiments for urban tree-crown detection, and develop reproducible notebooks, GIS outputs and an interactive research website. Public release: https://urbantreecrownmapping.catch4d.de/ Code and notebooks: https://gitlab.com/ils-research/catch4d-tree-detection-and-crown-volume.

ReVerSO — Student Research Assistant: automate spatial-analysis workflows in Jupyter notebooks and prepare GIS layers for a regional study of access to basic services in the Southern Upper Rhine region.

10/2025-01/2026

Research and Concept Development | CSCP / Catalyst Centre, Krefeld

Four-month business pilot project in an interdisciplinary team, in cooperation with KREFELD BUSINESS: an incentive-based circular rental model for the future Startup Centre Krefeld. Researched good practices, co-developed tenant commitments and incentives, supported stakeholder validation, and co-produced an agreement prototype and implementation roadmap. Certificate for active and outstanding participation.

2021-2023

Co-founder and Researcher | RohamRobotic, Tallinn

Developed a start-up concept through research, validation, planning and minimum viable product development in an interdisciplinary team.

2021-2022

Program Coordinator | Remolab, France (remote)

First programme coordinator of a pre-incubation programme for international start-up founders; helped set up and launch its first cycle. Planning, coordination and task tracking with Airtable and Miro.

Education

In progress

MSc Spatial Planning | TU Dortmund University

English-taught studies. Selected areas: urban transformation, European housing typologies and environmental justice in industrial regions. Thesis remains to be completed.

2016-2019

MSc Urban Planning | Iran University of Science and Technology, Tehran

Completed September 2019. Final grade: 18.43 on the Iranian 0-20 scale (20 = best, 12 = pass). National entrance-examination rank: 34 of 3,274 in urban and regional planning.

Cross-disciplinary development

2024-2025 **DLAB: Certificate in Digital Technology Management | TU Dortmund**
Three semesters, 33.5 ECTS. Two 7.5-ECTS electives: Theory and Methods in Management, Technology, and Leadership Research; Strategic Technology and Innovation Management. Further components: corporate digital innovation with Swisslog, lean start-up, venture creation, company workshop and practical experience.

2024 **Summer School for Sustainability | TU Dortmund**
Group work on the UN Sustainable Development Goals and the latest IPCC reports, closing with an assessed poster presentation.

2023 **Project management certificates**
Google Project Management Professional Certificate (Coursera); Career Essentials in Project Management (Microsoft and LinkedIn).

Selected analytical projects

Public repository **Dortmund Stadtregion Indicators**
Reproducible analysis of 33 municipalities using official regional indicators and municipal identifiers (AGS). Python notebooks, interactive mapping and a GIS data package. https://github.com/sophia-chehre/dortmund-stadtregion-indicators

2026 **Municipal Building Explorer | Independent project**
Comparison of official and open building datasets in selected areas of Bochum and Witten. Data harmonisation, coverage and geometry checks, interactive comparison and a documented GIS package. https://sophia-chehre.github.io/portfolio/projects/municipal-building-explorer/

Portfolio **Remote sensing and GeoAI workflows**
Implemented workflows for object detection and land-cover classification in ArcGIS, from imagery and training inputs through model execution, evaluation and mapped outputs. https://sophia-chehre.github.io/portfolio/projects/geoai-workflows/

Research outputs and publications

2026 **Chehreghan, Z., Pissulla, V. L. and Xu, S.**
CATCH4D Upscaling - tree detection and crown volume (Troisdorf), v1.0.2. ILS gGmbH. Zenodo. Published research package with data, code and documentation. https://doi.org/10.5281/zenodo.23734111

2022 **Ghorbanian, M. and Chehreghan, Z.**
The Relation between Urban Physical Indicators and Solar Irradiation Availability on Building Envelopes: Using Tehran as a Case Study. In Advances in Urbanism, Smart Cities, and Sustainability, pp. 227-239. CRC Press. Derived from MSc research completed in 2019. https://doi.org/10.1201/9781003126195-16

Methods and languages

Methods **Spatial analysis, programming and research**
Python, GeoPandas, pandas, Rasterio, QGIS/PyQGIS, ArcGIS Pro and Jupyter; raster/vector processing, data integration, automation, validation and reproducible documentation. Research design, regression analysis, stakeholder interviews, co-creation, roadmapping and interactive communication.

Languages **Persian | English | German**
Persian: native. English: professional working proficiency. German: B1 (certified).